

Sung Guk (John) Lee

Naju, South Korea · +82 10-3104-3723 · katsuya245126@gmail.com
sungguklee.com · github.com/katsuya245126 · linkedin.com/in/sungguklee
English (native) · Korean (conversational) · Japanese (conversational)

SUMMARY

Software engineer (B.S. Computer Science, Summa Cum Laude, 2026) who built and deployed an AI English speaking app for my own students: a TypeScript/Next.js codebase backed by 2,100+ automated tests. Completed the degree part-time over five years while teaching full-time.

SKILLS

Languages: TypeScript, JavaScript, Python, SQL

Frameworks: Next.js (App Router), React, Node.js, Flask, Jinja

Databases: PostgreSQL, Supabase, SQLite

AI / APIs: OpenAI (speech-to-text, TTS, LLM structured output, prompt design, moderation), Azure Speech

Testing: Vitest, Playwright end-to-end tests, pytest, CI/CD (GitHub Actions)

Infrastructure: Vercel, Supabase Auth & Storage, row-level security, SQL migrations, cron jobs, Git

AI tools: Claude Code, Codex

PROJECTS

Coco English

June 2026 – Present

TypeScript · Next.js 15 · React 19 · Node.js · Supabase/PostgreSQL · OpenAI · Azure Speech · Vercel

coco-english.vercel.app

AI English speaking practice app, live in production, built for and used by my own students. It replaced a manual loop where I listened to every recording and typed out corrections. The app transcribes each answer, checks meaning and target-pattern use, and returns pronunciation feedback.

- Cut feedback latency on exact-match answers from 3.5-13s to about 700ms with a deterministic fast path that skips the LLM call when the answer matches exactly.
- Evaluated a cheaper, faster model against real student transcripts and rejected the swap because it marked echoed questions as correct answers.
- 2,100+ automated tests plus Playwright end-to-end specs run on every push in a GitHub Actions CI/CD pipeline, with paid API adapters stubbed at the boundary.
- Built the frontend and backend with Claude Code and Codex, making the accept or reject call on each change: Next.js App Router on Node.js, 15 API endpoints, PostgreSQL via Supabase with 53 versioned SQL migrations, Vercel cron jobs.
- Built phoneme-level pronunciation scoring on Azure Speech with a per-student weakness profile, and validated the 60/40 accuracy/fluency weighting against 12 student recordings I scored by hand.

University IT Support Ticketing System — Course project

2025

Python · Flask · SQLite · Jinja · pytest

github.com/CSCI441Group5/academic-institution-ticketing-software

- Service desk ticketing system for schools and small university departments, where commercial platforms like Zendesk and ServiceNow are too complex and expensive. Replaces the shared inboxes and spreadsheets these teams fall back on.
- Users submit requests by category; the system routes each ticket to the right department queue, with separate dashboards for student, staff, and manager roles and filtering by category, department, and status.
- Primary contributor on a five-person team, authoring the majority of the backend and the automated test suite.

EXPERIENCE

ESL Teacher — Sprout English

Jan. 2022 – Present

Naju, South Korea

- Design and deliver English curriculum for elementary and middle school students across multiple grade levels and proficiency bands.
- Build lesson materials, assessments, and pacing systems, iterating based on measured student performance.

ESL Teacher — Ran English

2016 – 2018, 2021

Mandatory Military Service — Alternative Service, Republic of Korea

Jan. 2019 – Oct. 2020

EDUCATION

Fort Hays State University — B.S. Computer Science

May 2026

Summa Cum Laude · Dean's Honor List · Fort Hays Honor Society · Merit Scholarship Recipient